Assignment 2
IB3702 Mathematics for Machine learning

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November 7, 2025

Problem 1

exercise a

$Av = \Lambda v$

First I am going to multiply A to the first column of Q

So $v_1 = [-a, 0, b]^T$ with eigenvalue 25.

This gets me the equation:

$$A \begin{pmatrix} -a \\ 0 \\ b \end{pmatrix} = 25 \begin{pmatrix} -a \\ 0 \\ b \end{pmatrix}$$

Now I calculate $Av_1 =$

$$Av_1 = \begin{pmatrix} -2 & 0 & -36 \\ 0 & -3 & 0 \\ -36 & 0 & -23 \end{pmatrix} \begin{pmatrix} -a \\ 0 \\ b \end{pmatrix}$$

Now we multiply horizontally on the left, with vertically on the right. So for example the element on the top left will be $(-2)(-a)$ and then $0(0)$ and then $-36(b)$ and so on.

$$\begin{pmatrix} (-2)(-a) & +0(0) & +(-36)(b) \\ 0(-a) & +(-3)(0) & +0(b) \\ (-36)(-a) & +0(0) & +(-23)(b) \end{pmatrix} = \begin{pmatrix} 2a - 36b \\ 0 \\ 36a - 23b \end{pmatrix}$$

Now I equate this to $25v_1$:

$$\begin{pmatrix} 2a - 36b \\ 0 \\ 36a - 23b \end{pmatrix} = 25 \begin{pmatrix} -a \\ 0 \\ b \end{pmatrix} = \begin{pmatrix} 25a \\ 0 \\ 25b \end{pmatrix}$$

And now I can just solve the equations for a and b:

$$-2a - 36b = -25a \implies 27a - 36b = 0 \implies 27a = 36b \implies b = \frac{27}{36}a = \frac{3}{4}a$$

$$36a - 23b = 25b \implies 36a - 48b = 0 \implies 36a = 48b \implies b = \frac{36}{48}a = \frac{3}{4}a$$

Since Q orthogonally diagonalizes A, the following must be true

$$(-a)^2 + (0)^2 + (b)^2 = 1 \implies a^2 + b^2 = 1$$

I can enter the value of b I found above into this equation so we get:

$$a^2 + \left(\frac{3}{4}a\right)^2 = 1 \implies a^2 + \frac{9}{16}a^2 = 1 \implies \frac{25}{16}a^2 = 1 \implies a^2 = \frac{16}{25} \implies a = \frac{4}{5}$$

Then enter this value of a into the equation for b:

$$b = \frac{3}{4}a = \frac{3}{4} \cdot \frac{4}{5} = \frac{12}{20} = \frac{3}{5}$$

$$\boxed{a = \frac{4}{5}, b = \frac{3}{5}}$$

exercise b

To prove that Q is orthogonal, I will shows that the inverse of it is equal to its transpose

$$Q^{-1} = Q^T$$

First I calculate Q^T:

$$Q = \begin{pmatrix} -\frac{4}{5} & 0 & \frac{3}{5} \\ 0 & 1 & 0 \\ \frac{3}{5} & 0 & \frac{4}{5} \end{pmatrix} \to Q^T = \begin{pmatrix} -\frac{4}{5} & 0 & \frac{3}{5} \\ 0 & 1 & 0 \\ \frac{3}{5} & 0 & \frac{4}{5} \end{pmatrix}$$

Then I calculate Q^{-1} using the formula for the inverse of a 3×3 matrix:

$$Q^{-1} = \frac{1}{\det(Q)} adj(Q)$$

First I calculate the determinant of Q:

$$\det(Q) = -\frac{4}{5} \cdot \left(1 \cdot \frac{4}{5} - 0 \cdot 0\right) - 0 + \frac{3}{5} \cdot \left(0 \cdot 0 - 1 \cdot \frac{3}{5}\right) = -\frac{16}{25} - \frac{9}{25} = -1$$

Now I calculate the adjugate of Q:

$$adj(Q) = \begin{pmatrix} \begin{vmatrix} 1 & 0 \\ 0 & \frac{4}{5} \end{vmatrix} & -\begin{vmatrix} 0 & 0 \\ \frac{3}{5} & \frac{4}{5} \end{vmatrix} & \begin{vmatrix} 0 & 1 \\ \frac{3}{5} & 0 \end{vmatrix} \\ -\begin{vmatrix} 0 & \frac{3}{5} \\ 0 & \frac{4}{5} \end{vmatrix} & \begin{vmatrix} -\frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & \frac{4}{5} \end{vmatrix} & -\begin{vmatrix} -\frac{4}{5} & 0 \\ \frac{3}{5} & 0 \end{vmatrix} \\ \begin{vmatrix} 0 & \frac{3}{5} \\ 1 & 0 \end{vmatrix} & -\begin{vmatrix} -\frac{4}{5} & \frac{3}{5} \\ 0 & 0 \end{vmatrix} & \begin{vmatrix} -\frac{4}{5} & 0 \\ 0 & 1 \end{vmatrix} \end{pmatrix}$$

Calculating the determinants of the 2×2 matrices:

$$Cof(Q) = \begin{pmatrix} \frac{4}{5} & 0 & -\frac{3}{5} \\ 0 & 1 & 0 \\ -\frac{3}{5} & 0 & -\frac{4}{5} \end{pmatrix} \xrightarrow{adj(Q)=Cof(Q)^T} adj(Q) = \begin{pmatrix} \frac{4}{5} & 0 & -\frac{3}{5} \\ 0 & 1 & 0 \\ -\frac{3}{5} & 0 & -\frac{4}{5} \end{pmatrix}$$

Now I can calculate Q^{-1}:

$$Q^{-1} = \frac{1}{\det(Q)} adj(Q) = \frac{1}{-1} \begin{pmatrix} \frac{4}{5} & 0 & -\frac{3}{5} \\ 0 & 1 & 0 \\ -\frac{3}{5} & 0 & -\frac{4}{5} \end{pmatrix} = -1 \cdot \begin{pmatrix} \frac{4}{5} & 0 & -\frac{3}{5} \\ 0 & 1 & 0 \\ -\frac{3}{5} & 0 & -\frac{4}{5} \end{pmatrix} = \begin{pmatrix} -\frac{4}{5} & 0 & \frac{3}{5} \\ 0 & 1 & 0 \\ \frac{3}{5} & 0 & \frac{4}{5} \end{pmatrix}$$

$$\boxed{\text{Since } Q^{-1} = Q^T\text{, } Q \text{ is orthogonal.}}$$

exercise c

Two vectors are orthogonal if their dot product is 0, so I will just have to calculate the dot product for each column vector pair in Q.

$$v_1 = \begin{pmatrix} -\frac{4}{5} \\ 0 \\ \frac{3}{5} \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} \frac{3}{5} \\ 0 \\ \frac{4}{5} \end{pmatrix}$$

Calculating $v_1 \cdot v_2$:

$$v_1 \cdot v_2 = \left(-\frac{4}{5}\right)(0) + (0)(1) + \left(\frac{3}{5}\right)(0) = 0$$

Calculating $v_1 \cdot v_3$:

$$v_1 \cdot v_3 = \left(-\frac{4}{5}\right)\left(\frac{3}{5}\right) + (0)(0) + \left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = -\frac{12}{25} + \frac{12}{25} = 0$$

Calculating $v_2 \cdot v_3$:

$$v_2 \cdot v_3 = (0)\left(\frac{3}{5}\right) + (1)(0) + (0)\left(\frac{4}{5}\right) = 0$$

$$\boxed{\text{Since all dot products are 0, the column vectors of } Q \text{ are pairwise orthogonal.}}$$

exercise d

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exercise e

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Problem 2

exercise a

Like mentioned in the hint, I will first show linear independence and then show that they span the space.

Linear independence

To show that the set of vectors is linearly independent, I need to show that the only solution to the equation:

$$c_1v_1 + c_2v_2 + c_3v_3 = 0$$

is the trivial solution where all coefficients are 0.

For us the equation becomes:

$$c_1(-1 + x) + c_2(x + x^2) + c_3(1 + x + x^2) = 0$$

So I set up the equation:

$$c_1(-1) + c_1(x) + c_2(x) + c_2(x^2) + c_3(1) + c_3(x) + c_3(x^2) = 0$$

This gives me the following system of equations:

$$\begin{cases} -c_1 + c_3 = 0 \\ c_1 + c_2 + c_3 = 0 \\ c_2 + c_3 = 0 \end{cases}$$

From the first equation, I deduce that $c_3 = c_1$. A numerical example $-(7) + 7 = 0$ shows that this is correct.

Then I substitute c_3 in the second equation:

$$c_1 + c_2 + c_3 = 0 \implies 2c_1 + c_2 = 0 \implies c_2 = -2c_1$$

Then I substitute c_2 and c_3 in the third equation:

$$c_2 + c_3 = 0 \implies -2c_1 + c_1 = 0 \implies -c_1 = 0 \implies c_1 = 0$$

Then substituting c_1 back to find c_2 and c_3:

$$c_2 = -2c_1 = -2(0) = 0$$

$$c_3 = c_1 = 0$$

So the only solution is $c_1 = c_2 = c_3 = 0$, which is the trivial solution. Therefore the set of vectors is linearly independent.

Spanning the space

A basis for $\mathbb{P}_2(\mathbb{R})$ must have 3 linearly independent vectors, because the dimension of $\mathbb{P}_2(\mathbb{R})$ is 3.

We have exactly 3 vectors, and have proven above that they are linearly independent. Therefore they also span the space $\mathbb{P}_2(\mathbb{R})$.

$$\boxed{\{\,-1+x, x+x^2, 1+x+x^2\}\text{ is indeed a basis for }\mathbb{P}_2(\mathbb{R}).}$$

exercise b

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exercise c

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exercise d

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exercise e

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Problem 3

exercise a

$$\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix} \xrightarrow{\text{reflection across } y=x} \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix} \xrightarrow{\text{reflection across } x\text{-axis}} \begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}$$

The combined matrix is the product of the two matrices. I call the matrix after the first reflection A and the one after the second reflection B:

$$BA = \begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}\begin{pmatrix}0 & 1\\1 & 0\end{pmatrix} = \begin{pmatrix}0 & 1\\-1 & 0\end{pmatrix}$$

Then if we multiply this with the original vector $\begin{pmatrix}x\\y\end{pmatrix}$ we get:

$$\begin{pmatrix}0 & 1\\-1 & 0\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}(0)(x)+(1)(y)\\(-1)(x)+(0)(y)\end{pmatrix} = \begin{pmatrix}y\\-x\end{pmatrix}$$

So this transformation is a 90 degree rotation clockwise

exercise b

1. The matrix B will be:

$$B = \begin{pmatrix}\cos(75°) & -\sin(75°)\\ \sin(75°) & \cos(75°)\end{pmatrix}$$

2.

- **Kernel of** B: a rotation by 75° will only map the zero vector to the zero vector, so the kernel of B only contains the zero vector. So the kernel of $B = \{0\}$
- **Image of** B: A rotation is invertible and maps $\mathbb{R}^2$ to $\mathbb{R}^2$, so the image of $B = \mathbb{R}^2$

3. B^{12} means we do the rotation 12 times. $12 \times 75° = 900°$. And knowing that a full rotation is 360°, the actual rotation is $(900 \mod 360) = 180°$. Therefore B^{12} is equivalent to a rotation of 180°.

$$B^{12} = \begin{pmatrix}\cos(180°) & -\sin(180°)\\ \sin(180°) & \cos(180°)\end{pmatrix} = \begin{pmatrix}-1 & 0\\0 & -1\end{pmatrix}$$

4. The matrix of B^{-1} is the matrix that rotates by $-75°$.

$$B^{-1} = \begin{pmatrix}\cos(-75°) & -\sin(-75°)\\ \sin(-75°) & \cos(-75°)\end{pmatrix} = \begin{pmatrix}\cos(75°) & \sin(75°)\\ -\sin(75°) & \cos(75°)\end{pmatrix}$$

Since rotations are orthogonal, $B^{-1} = B^T$

Problem 4

I construct the following matrix:

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 5 \end{pmatrix}$$

exercise a

I chose an upper triangular matrix, because in this type the determinant is easy to calculate, it's just the multiplication of the diagnonal elements.

Here: $\det(A) = 1 \cdot 1 \cdot 5 = 5$

And I also made sure to use simple numbers so the calculations would be easier.

exercise b

(10) True, because the determinant is 5 (calculated in exercise a), $\neq 0$

(1) True, because a square matrix is invertible if the determinant is nonzero (our determinant is 5)

(7)

To calculate the RREF of A I perform the following operations:

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 5 \end{pmatrix} \xrightarrow{R_3 = \frac{1}{5} \cdot R_3} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_2 = R_2 - 3R_3} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1 = R_1 - 2R_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \mathbb{I}_3$$

(3) True, because the null space only contains the zero vector when the RREF is the identity matrix, which we have shown above.

(4) True. From (3) we know that the null space only contains the zero vector, so the only solution to $A\vec{x} = \vec{0}$ is the trivial solution $\vec{x} = \vec{0}$

(6) The rank of a matrix is the number of pivot columns in the RREF and ours has 3 pivots, so $rank(A) = 3 = n$

(2) To prove this, we need to first fine the existence of a solution, and then prove the uniqueness of it.

<u>Existence of solution</u>

Since A is invertible (see (1)), we can multiply both sides of the equation $A\vec{x} = \vec{b}$ with A^{-1}:

$$A^{-1}A\vec{x} = A^{-1}\vec{b} \implies \vec{x} = A^{-1}\vec{b}$$

So there exists a solution for each $\vec{b} \in \mathbb{R}^n$

Uniqueness of solution

If I assume there are two solutions $\vec{x_1}$ and $\vec{x_2}$ so that:

$$A\vec{x_1} = \vec{b} \quad and \quad A\vec{x_2} = \vec{b}$$

Then that would mean that:

$$A(\vec{x_1} - \vec{x_2}) = \vec{0}$$

From (4) we know that the only solution to this equation is the trivial solution $\vec{x_1} - \vec{x_2} = \vec{0}$, so $\vec{x_1} = \vec{x_2}$

(5)

- (5.1) True, the columns are linearly independent when the RREF is the identity matrix (see above)
- (5.2) True, because the columns of A span $\mathbb{R}^n$ when A is invertible (see (1))

(8) The transpose of A is:

$$A^T = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 3 & 5 \end{pmatrix}$$

The determinant of A^T is also 5 ($1 \times 1 \times 5 = 5$), and a square matrix is invertible when the determinant is nonzero, so A^T is invertible.

(9) The rows of A are the columns of A^T, and since I showed in (5) that the columns of a matrix form a basis for $\mathbb{R}^n$ when the matrix is invertible, the rows of A also form a basis for $\mathbb{R}^n$

Problem 5

I think Problem 4 from assignment 1 was pretty cool since it was very big, and finally solving it completely felt very rewarding. The problem itself was only a few lines, but my solution spanned 3 pages. Granted the way I wrote up solutions in that assignment was very (I'll call it) 'verbose' so that might be why it spanned 3 pages ☺